High-burnup structure formation as a second-order phase transition

A Landau functional for SCIANTIX

`iHighBurnupStructureFormation = 4`

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Outline

SCIANTIX status

The model

Results

Model applied to SCIANTIX

SCIANTIX status

SCIANTIX had three formation models for the restructured volume fraction

option	form	driven by
1	KJMA, Barani (2020)[1]	effective burnup
2	KJMA + $bu_{\mathrm{inc}} = 15$, Biswas & Aagesen (2025)[2]	effective burnup
3	KJMA on ρ_d, Veshchunov & Shestak (2009)[3]	local burnup

All three are transformation kinetics: a sigmoid fitted in burnup. None of them produces the **mean misorientation** Θ of the subgrains, or the **subgrain radius** r_n.

[1] Barani *et al.*, *J. Nucl. Mater.* 539 (2020) 152296.
[2] Biswas & Aagesen, *Comput. Mater. Sci.* 258 (2025) 114052, Eq. 45.
[3] Veshchunov & Shestak, *J. Nucl. Mater.* 384 (2009) 12–18.

What KJMA is, and what it assumes

Kolmogorov, Johnson & Mehl and Avrami, independently, 1937–41.[4] Nuclei appear *at random* in the untransformed material and grow *isotropically*; count the transformed volume as if the growing regions could overlap freely — the extended volume V_e — and the overlap correction is purely statistical:

$$\mathrm{d}X = \left(1 - X\right) \frac{\mathrm{d}V_e}{V} \quad \Longrightarrow \quad X = 1 - \exp\left(-K\,t^{\,n}\right)$$

What you get

Two numbers, K and n, and a sigmoid. n carries the dimensionality of growth and the nucleation regime — but only if the assumptions hold.

What it assumes

Randomly placed nuclei, a growth law independent of where you are in the transformation, and a *clock*. In SCIANTIX the clock is burnup.

[4] Kolmogorov (1937); Johnson & Mehl, *Trans. AIME* 135 (1939) 416; Avrami, *J. Chem. Phys.* 7 (1939) 1103. Derivation: Christian (2002), ch. 12.

The three options are three answers to “which clock?”

1. **Effective burnup**, $n = 3.54$, $K = 2.77 \times 10^{-7}$, fitted on restructured-fraction data. The transformation begins the instant irradiation does.
2. Same, with an **incubation burnup** $bu_{\text{inc}} = 15\,\text{MWd/kgHM}$ below which nothing happens. Biswas & Aagesen get it from a balance between the energy stored in dislocations and the energy of forming a subgrain — the same physical argument as this work, used to shift the origin of an empirical curve rather than to build the curve.
3. KJMA not on burnup but on the **dislocation density** $\rho_d(bu, T)$, with a threshold $\rho_{\text{crit}} = 6 \times 10^{14}\,\text{m}^{-2}$. The first to make the driver a **microstructural state variable** instead of a clock, and the only one of the three that feels temperature.

What none of them can do

They are calibrated on X alone. There is no Θ, no r_n, and no internal state: ask the model where the microstructure *is* and it can only tell you how far along a fitted curve it has travelled.

Why a phase transition is the right shape for this problem

The HBS forms *continuously*

Not by nucleating new grains, but by low-angle boundaries absorbing dislocations until their misorientation drifts through the 10° HAGB threshold. In the metallurgical literature this is *continuous* dynamic recrystallization, and it is a well-posed alternative to the nucleation-and-growth picture KJMA describes.[5]

The evidence in UO_2 is the EBSD itself: Θ rises smoothly with burnup. There is no population of already-10° nuclei waiting to grow.

A continuous transformation has an order parameter

If a single continuous quantity distinguishes the two states, Landau's argument applies: expand the free energy in it, and keep only the terms symmetry allows.[6]

$$F(\eta) = C_0 + C_2\eta^2 + C_4\eta^4$$

Odd powers are forbidden here for a concrete reason — the energy of a boundary cannot depend on the *sign* of the misorientation across it.

C_2 changing sign *is* the transition. Nothing is fitted to make it happen.

[5] Gourdet & Montheillet, *Acta Mater.* 51 (2003) 2685; Humphreys & Hatherly (2017), chs. 6 and 10.

[6] Landau (1937); Landau & Lifshitz, *Statistical Physics* I, §143.

Landau against KJMA, side by side

	KJMA (options 1–3)	Landau (option 4)
what is solved	a *kinetic* law along a path	the *equilibrium* of a state
driver	a clock (burnup), or ρ_d	$\rho_{\mathrm{tot}}(bu)$ through C_2
free parameters	K, n fitted on X	β, k, ρ_c fitted on Θ, r_n
predicts X	by construction	**as a by-product**
predicts Θ, r_n	no	**yes**
reversible?	no, monotone by construction	yes — see below

The price of using an equilibrium theory

F is minimised afresh at every step, so a drop in ρ_{tot} would un-restructure the fuel. Restructuring is irreversible, so the implementation carries a monotonic lock on Θ, and re-derives r_n and X from the locked value. It is the one piece of the model that is kinetics rather than thermodynamics.

The data this model is built on

EBSD — Zacharie-Aubrun *et al.* (2022), Onofri *et al.* (2025). 42 radial positions on 7 rods, with TRANSURANUS local conditions.

quantity	N	what it is
mean misorientation Θ	41	$[\,A_{\mathrm{Mis}}(f_1 - f_{10}) + 10 f_{10}\,]/100$
restructured fraction X	27	f_{10}, area fraction above 10°
subgrain radius r_n	14	ECD50 % /2

The model

HBS as a second-order phase transition

Order parameter

$$\eta = \frac{\theta}{\theta_{\text{max}}}, \qquad \theta_{\text{max}} \equiv \theta_{\text{HAGB}} = 10^\circ$$

the mean misorientation normalised to the LAGB/HAGB boundary.

External condition: the local burnup, entering *only* through the dislocation density

$$\rho_{\text{tot}}(bu) = 10^{\,2.2\times10^{-2}\,bu\,+\,13.8}$$

Nogita & Une (1994)

Free energy

$$F(bu,\eta) = C_0 + C_2\,\eta^2 + C_4\,\eta^4$$

Only even powers: the energy cannot depend on the *sign* of a misorientation.

Minimising F gives Θ; the same wall geometry gives r_n; the lever rule gives X.

Three adjustable parameters

β (wall geometry), k (sweeping), ρ_c (strain-field cut-off).

The geometric wall

A low-angle boundary *is* a wall of dislocations. Two classical relations fix the whole geometry.

1. Spacing sets the angle $\theta = b/d$

so a wall of n dislocation families carries a line length per unit *wall area*

$$L_A = n\theta/b$$

2. Wall area per unit volume

$$S/V = 3\sqrt{\rho_{\mathrm{LAGB}}}\,/\,\beta$$

β: the one geometric parameter (Rest & Hofman 2000).
$n = 2$, Gourdet & Montheillet, not fitted (between 1-3).

Close the two on each other:

$\rho_{\mathrm{LAGB}} = L_A \cdot S/V$

$$\implies \quad \sqrt{\rho_{\mathrm{LAGB}}} = \frac{3n\theta}{\beta b}$$

Everything follows from θ alone

$$\rho_{\mathrm{LAGB}} = \left(\frac{3n\theta}{\beta b}\right)^2, \qquad r_n \simeq \frac{\beta^2 b}{6n\theta}$$

The dislocation balance

Every dislocation is in exactly one of three states, and they must add up to ρ_{tot}:

$$\underbrace{\rho_{\text{ord}} = \rho_{\text{LAGB}}^{\max}\,\eta^2}_{\text{condensed into low-angle walls}} \;+\; \underbrace{\rho_{\text{swept}} = (\rho_{\text{tot}} - \rho_{\text{ord}})\,\frac{\Delta R}{R}}_{\text{annihilated by moving boundaries}} \;+\; \underbrace{\rho_{\text{free}}}_{\text{still a random array}} \;=\; \rho_{\text{tot}}$$

Each population carries the line energy of the state it is in:

$$F \;=\; \rho_{\text{free}}\,A_1\,Gb^2 \;+\; \rho_{\text{ord}}\,A_2\,Gb^2, \qquad A_i = \frac{f(\nu)}{4\pi}\ln\frac{R_i}{b}$$

free ones screened at ρ_c, wall ones by the wall itself at ρ_{tot}; the swept ones carry nothing.

The functional, collected

$$C_0 = \rho_{\text{tot}} A_1 G b^2, \qquad C_4 = \rho_{\text{LAGB}}^{\max} \frac{\Delta R}{R}\Big|_{\max} A_1 G b^2$$

$$C_2 = \underbrace{\rho_{\text{LAGB}}^{\max} (A_2 - A_1)\, G b^2}_{\text{walls: the screening gain}} \underbrace{- \rho_{\text{tot}} \frac{\Delta R}{R}\Big|_{\max} A_1 G b^2}_{\text{sweep}}$$

The engine of the transition is $A_2 - A_1$. As burnup rises, ρ_{tot} rises, the screening length inside the wall shrinks, A_2 falls below A_1 — and it becomes cheaper to store dislocations in walls than to leave them random.

Equilibrium

$$\frac{\partial F}{\partial \eta} = 0 \implies \eta^2 = -\frac{C_2}{2C_4}, \quad \text{clipped at 0}$$

$\eta = 0$ wherever $C_2 \geq 0$.

Condition on the minimisation problem: dislocation balance

The walls cannot hold more dislocations than exist, so $\rho_{\text{ord}} \le \rho_{\text{tot}}$. The equilibrium is the minimum of F on the admissible interval, not the free stationary point:

Eq. (7b) — admissibility

$$\eta_{\text{bal}} = \sqrt{\min\left(\frac{\rho_{\text{tot}}}{\rho_{\text{LAGB}}^{\text{max}}},\ 1\right)}, \qquad \eta_{\text{eq}} = \min\left(\sqrt{-\frac{C_2}{2C_4}},\ \eta_{\text{bal}}\right)$$

Results

Calibration

Joint fit of β, k, ρ_c on Θ *and* the subgrain sizes, while the restructured fraction is *not* in the objective.

$$J = \frac{\langle \Delta\Theta^2 \rangle}{\mathrm{var}\,\Theta} + w\,\frac{\langle \Delta r^2 \rangle}{\mathrm{var}\,r}$$

dimensionless and symmetric; `differential_evolution`, 6 seeds, all converging to $J = 0.2398$.

n (not fitted)	2
β	33.547
k	0.04697
ρ_c	1.166e15 m^{-2}

Three parameters, two datasets, three predictions

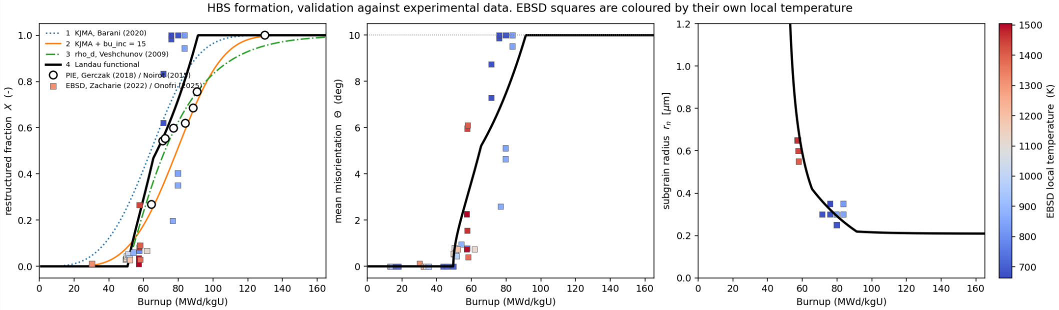
β, k and ρ_c are fitted on Θ and r_n only. The restructured fraction X — the one quantity SCIANTIX actually consumes — is left out of the objective and comes out of the lever rule, Eq. (10), as a **prediction**.

That is what makes the next slide a test rather than a report of the fit: X and the PIE fractions are scored on data the calibration never saw.

What the balance bought

		RMSE / R^2
Θ	$(N = 41$, calibrated)	1.7616° / 0.772
X	$(N = 27$, *not* fitted)	**0.2122 / 0.710**
r_n	$(N = 14$, calibrated)	0.0678 μm / 0.774
PIE fraction ($N = 8$, out-of-sample)		0.1461 / +0.433

Validation against both datasets



Against the three models SCIANTIX already had

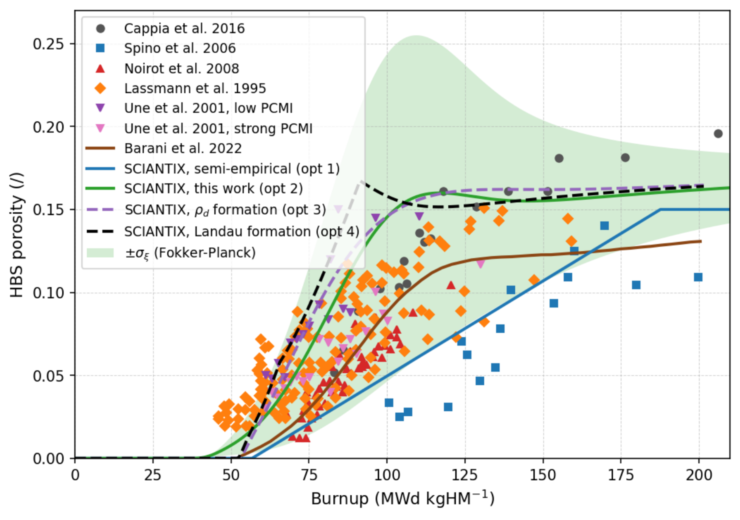
Same history, same initial conditions, same time stepping; only the formation model varies.
Burnup [MWd/kgU] at which α_r first reaches:

threshold	1 KJMA	2 KJMA+bu_{inc}	3 ρ_d	**4 Landau**
1 %	19.5	34.5	51.2	**51.1**
10 %	37.8	52.8	54.8	**53.5**
50 %	64.2	79.3	72.3	**67.7**
90 %	90.1	105.2	112.3	**87.7**
99 %	109.7	124.6	161.7	**91.1**

- At the onset, option 4 lands **within 0.1 MWd/kgU of option 3** — an independently calibrated model built on a different microstructural variable, so a genuine cross-check and not a shared fit.
- They part at the top: the lever rule saturates when Θ hits the 10° cap, whereas KJMA approaches 1 asymptotically. Option 4 is done by 91 MWd/kgU, option 3 not before 162.
- Option 4 spans the transition in **40 MWd/kgU** against 90 for option 1: once the driver is a state variable rather than a fitted sigmoid in burnup, the transformation is sharper.

Model applied to SCIANTIX

Does the new formation model disturb the porosity model?



Largest deviation $+0.044$ **absolute at 91.3 MWd/kgHM** ($+36\,\%$), exactly the saturation burnup. Elsewhere: ≤ 0.026 below 75, ≤ 0.008 above 120, and the two end states are **0.7 % apart**.

Held against option 2 in `test_UO2HBS`; all four options end within **0.17 %** of the same porosity.

Summary

References i

[1] L.D. Landau, *On the theory of phase transitions*, Zh. Eksp. Teor. Fiz. **7** (1937) 19; L.D. Landau, E.M. Lifshitz, *Statistical Physics*, Part 1, 3rd ed., §143.

[2] A.N. Kolmogorov, Izv. Akad. Nauk SSSR, Ser. Mat. **1** (1937) 355–359.

[3] W.A. Johnson, R.F. Mehl, Trans. AIME **135** (1939) 416–442.

[4] M. Avrami, *J. Chem. Phys.* **7** (1939) 1103–1112; **8** (1940) 212–224; **9** (1941) 177–184.

[5] J.W. Christian, *The Theory of Transformations in Metals and Alloys*, Part I, 3rd ed., Pergamon (2002), ch. 12, *Formal Theory of Transformation Kinetics*.

[6] F.J. Humphreys, *A unified theory of recovery, recrystallization and grain growth, based on the stability and growth of cellular microstructures — I. The basic model*, *Acta Mater.* **45** (1997) 4231–4240.

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[7] F.J. Humphreys, G.S. Rohrer, A.D. Rollett, *Recrystallization and Related Annealing Phenomena*, 3rd ed., Elsevier (2017), ch. 6 (recovery) and ch. 10 (growth and stability of cellular microstructures).

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[10] J.F. Nye, *Some geometrical relations in dislocated crystals*, *Acta Metall.* **1** (1953) 153–162.

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[14] T. Barani *et al.*, *Modeling high burnup structure in oxide fuels... Part I: HBS formation*, *J. Nucl. Mater.* **539** (2020) 152296. *option 1*

[15] T. Barani *et al.*, *... Part II: Porosity evolution*, *J. Nucl. Mater.* **563** (2022) 153627. *porosity options 2–3*

[16] S. Biswas, L.K. Aagesen, *Comput. Mater. Sci.* **258** (2025) 114052, Eq. 45. *option 2*, bu_{inc}

[17] M.S. Veshchunov, V.E. Shestak, *Model for evolution of crystal defects in UO_2 under irradiation up to high burn-ups*, *J. Nucl. Mater.* **384** (2009) 12–18. *option 3, $\rho_d(bu, T)$*

[18] G. Khvostov *et al.*, *Approaches to modeling of high burn-up structure... in the START-3 code*, WRFPM-2005, Kyoto.

[19] K. Nogita, K. Une, *Radiation-induced microstructural change in high burnup UO_2 fuel pellets*, *Nucl. Instrum. Methods B* **91** (1994) 301–306. *Eq. (1), $\rho_{tot}(bu)$*

[20] K. Nogita, K. Une, *Irradiation-induced recrystallization in high burnup UO_2 fuel*, *J. Nucl. Mater.* **226** (1995) 302–310.

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[22] J. Spino *et al.*, *J. Nucl. Mater.* **354** (2006) 66–84. *porosity option 1, and the swelling data*

[23] I. Zacharie-Aubrun *et al.*, *Restructuring in high burn-up UO_2 fuels: experimental characterization by electron backscattered diffraction*, *J. Appl. Phys.* **132** (2022) 195903.
28 of the 42 EBSD points

[24] C. Onofri *et al.*, *Experimental characterizations by EBSD and TEM of sub-grain boundaries and dislocations in low irradiated UO_2 fuels*, *J. Nucl. Mater.* **615** (2025) 155981. *the other 14*

[25] NEA, *Recommendations on nuclear fuel properties*, NEA/NSC/R(2024)1 (2025), p. 124.
Eq. (2), G and ν

[26] D. Pizzocri *et al.*, *SCIANTIX: a new open source multi-scale code for fission gas behaviour modelling*, *J. Nucl. Mater.* **532** (2020) 152042.

[27] G. Zullo *et al.*, *The SCIANTIX code for fission gas behaviour: status, upgrades, separate-effect validation and future developments*, *J. Nucl. Mater.* **587** (2023) 154744.

Backup — the equations

(1) $\rho_{\mathbf{tot}}(bu) = 10^{2.2\times 10^{-2} bu + 13.8}$

(2) $G(T, P, x, q),\ \nu(T, P, x, q)$ NEA/NSC/R(2024)1 p. 124; $f(\nu) = \frac{1-\nu/2}{1-\nu}$

(3) $\rho^{\max}_{\mathbf{LAGB}} = \left(\frac{3n\theta_{\max}}{\beta b}\right)^2, \quad (S/V)_{\max} = \frac{9n\theta_{\max}}{\beta^2 b}, \quad \left.\frac{\Delta R}{R}\right|_{\max} = \frac{k\,\rho^{\max}_{\mathbf{LAGB}}}{\rho_{\mathbf{tot}}}$

(4) $A_1 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_c^{-1/2}}{b}, \qquad A_2 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\mathbf{tot}}^{-1/2}}{b}$

(5,6) $F = \rho_{\mathbf{free}} A_1 G b^2 + \rho_{\mathbf{ord}} A_2 G b^2 = C_0 + C_2 \eta^2 + C_4 \eta^4$

(7,7b) $\eta_{\mathbf{eq}} = \min\left(\sqrt{-C_2/2C_4},\ \eta_{\mathbf{bal}}\right), \quad \eta_{\mathbf{bal}} = \sqrt{\min(\rho_{\mathbf{tot}}/\rho^{\max}_{\mathbf{LAGB}}, 1)}$

(8) $\Theta = \min(\eta_{\mathbf{eq}} \theta_{\max} \cdot \frac{180}{\pi},\ \theta_{\mathbf{HAGB}})$

(9) $r_n = \min\left(\frac{1.5}{(S/V)}(1 + \frac{\Delta R}{R}),\ R_{\mathbf{grain}}\right)$

(10) $X = \mathrm{clip}\left(\frac{\Theta - \theta_u}{\theta_{\mathbf{HAGB}} - \theta_u},\ 0,\ 1 - 10^{-9}\right)$

Backup — regimes and parameters

Regimes (T-, P-, x-independent)

condition	bu [GWd/tU]
$C_2 = 0$, Θ leaves 0	49.56
$\Theta = \theta_u$, X leaves 0	50.81
$\Theta = \theta_{\mathrm{HAGB}}$	91.32

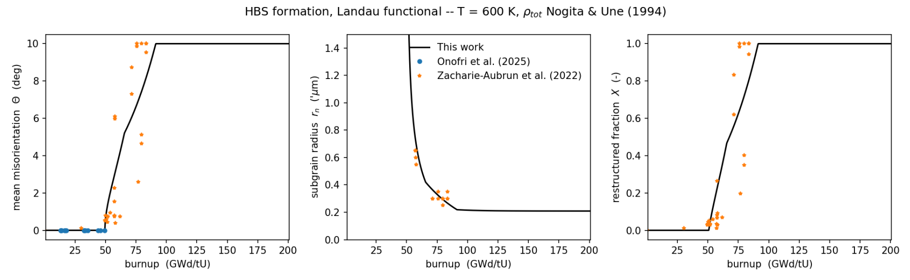
The balance bound sets η over roughly **65.6–91.3 GWd/tU**, and at 11 of the 41 EBSD points.

Fixed constants

b	3.889e-10 m
$\theta_{\max}$	$= \theta_{\mathrm{HAGB}}$
θ_{HAGB}	10°
θ_u	1.00°
n	2

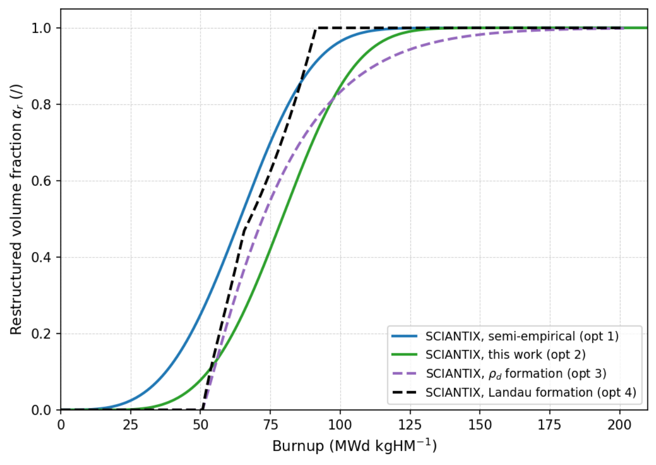
θ_u is the misorientation of the unrestructured matrix, read off the data rather than fitted: the measured A_{Mis2Mean} spans 0.1–6.1° with a median of 1.5°, and 1.0° is the round value taken inside that range. It sets only *where* X leaves zero, not the shape of the transition: Θ and r_n do not depend on it.

Backup — the model on its own, $T = 600\,\mathrm{K}$, $P = 0.05$




```
python3 hbs_formation_landau.py --plot
```

Backup — the four regression cases, as they ship



Not the controlled comparison of the previous figure: each case pairs its formation option with the porosity model it was built around (1, 2, 3, 3), so part of the spread is the porosity model.

This is what a user of each regression case actually gets; `formation_options.png` is what isolates the formation model.

Backup — the four formation options in a full run